

SHD training smoke test — it trains

exp060 · 11 July 2026 · Draft

What this checks

Before the program plan’s baseline can be trusted, one thing has to hold: the SHD training pipeline actually **trains**. This is that check — deliberately a low bar. It runs Rung A of the plan, the free signed-recurrent ceiling (all four recurrent blocks trainable, *no-dale-law*, gamma untuned), on a small subset for a few dozen epochs, and asks only whether the loss falls and test accuracy climbs off the 5% chance floor. It is engineering validation, *not* a registered result — the baseline number the plan pins comes from a full-scale run.

Method

Parameter	Value
Dataset	SHD, 1000-sample subset (of 8156)
Model	COBANet, 256 hidden, all four recurrent blocks trainable, signed (no Dale’s law)
Integration	$\Delta t = 1.0$ ms, $T = 1000$ ms ($T_{\text{steps}} = 1000$) — coarse dt so a full 1 s window still trains in minutes
Training	30 epochs, lr 0.001, batch 32, surrogate-gradient BPTT
Regulariser	upper firing-rate bound (Cramer et al. SHD values, $\theta_u = 100$, $s_u = 0.06$) — without it the signed recurrence runs away

The coarse $\Delta t = 1$ ms is the fastest our conductance model tolerates (verified on MNIST); it is still $\approx 10\times$ finer than Cramer et al.’s discrete-LIF timestep, because our fast synaptic and inhibitory time constants ($\tau_{\text{AMPA}} = 2$ ms, $\tau_{m,I} = 5$ ms) need resolving. Where:

- Δt , the integration timestep;
- $T_{\text{steps}} = T/\Delta t$, the number of BPTT steps unrolled per utterance.

Results

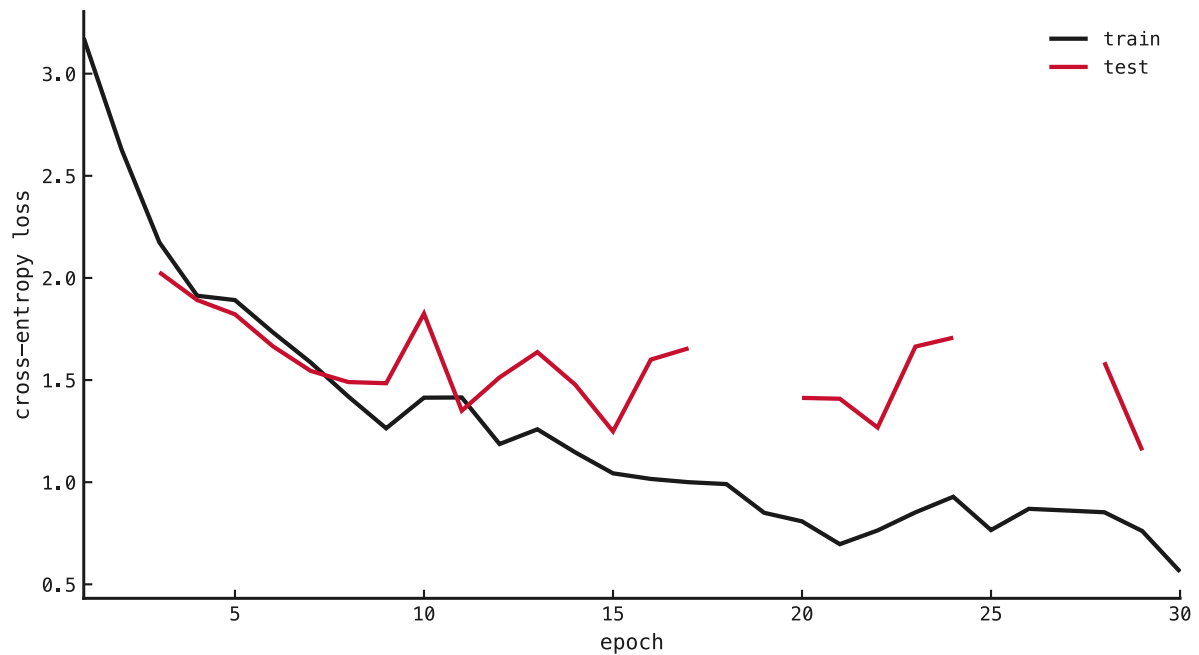


Figure 1: Cross-entropy loss against epoch. **What we expect.** If the pipeline learns, training loss falls. **What we see.** It does — training loss drops from 3.17 to 0.56. Test loss falls with it for the first few epochs, then turns upward: the network is overfitting the 1000-sample subset, exactly as a small training set predicts.

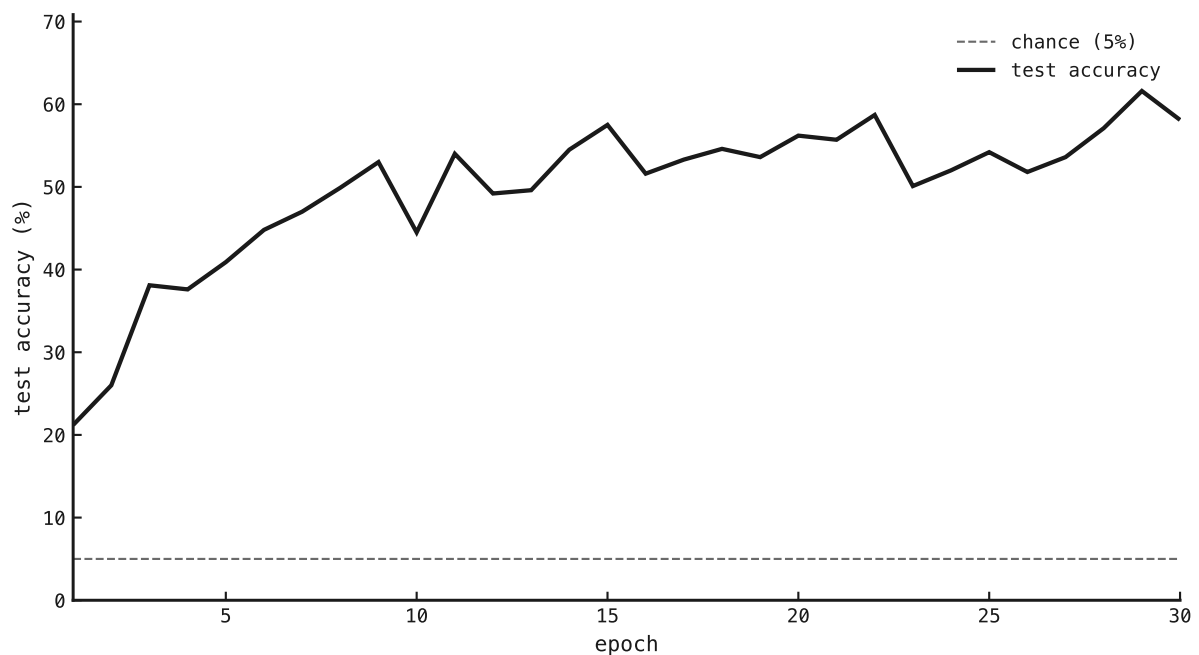


Figure 2: Test accuracy against epoch, with the 5% chance floor (20 classes) drawn in. **What we see.** Accuracy climbs from 21% at epoch 1 to a best of 62% at epoch 29 — an order of magnitude above chance — then plateaus and wobbles (final 58%) as the subset overfits.

Reading

Both bars are cleared: the loss falls and accuracy reaches 62%, far above both the 5% floor and the plan’s “decisively above chance” threshold of 25%. So the SHD data path, the binning, the

all-four-trainable signed recurrence, and the surrogate-gradient training loop all work end to end. The firing-rate regulariser is load-bearing: without it the earlier unregularised run blew up to a 526 Hz inhibitory rate; with it, rates stay physiological and the network learns.

The overfitting is the expected cost of the small subset and is *why* this is a smoke test, not a result: the honest ceiling number has to come from a full-scale run, where the test set is not swamped by a 1000-sample training set. That full-scale run is the plan’s registered Rung A. Next: scale the subset up (and add mild early-stopping) for the number, then descend the ladder to the Dale’s-law-constrained baseline.